COS 201

COMPUTER PROGRAMMING I

Assignment Answers — Weeks 1, 2 and 3

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Programme	Computer science
Course	COS 201 — Computer Programming I

Prepared from the COS 201 lecture material for Weeks 1–3. Replace the student-information blanks and sample details before submission.

WEEK 1: INTRODUCTION TO PROGRAMMING AND PROBLEM-SOLVING

Activity A — Identify the IPO Components

1. Calculate the area of a rectangle (length = 12, width = 5).

Input: Length = 12, Width = 5.
Process: Area = Length × Width = 12 × 5.
Output: 60 square units.

2. Calculate the average of four examination scores.

Input: Four examination scores.
Process: Add the four scores and divide by 4.
Output: Average score.

3. Determine whether a number is positive, negative or zero.

Input: A number.
Process: Check whether it is greater than, less than or equal to zero.
Output: Positive, Negative or Zero.

4. Calculate the total price of six items costing ■250 each.

Input: Quantity = 6, Price = ■250.
Process: 6 × 250.
Output: ■1,500.

5. Calculate the number of failed login attempts in a security log.

Input: Security log records.
Process: Identify and count all failed login attempts.
Output: Total number of failed login attempts.

Activity B — Write Algorithms

1. Algorithm to accept two numbers and display their sum.

1. Start.
2. Input the first number.
3. Input the second number.
4. Add both numbers.
5. Display the sum.
6. Stop.

2. Algorithm to calculate the area of a circle.

1. Start.
2. Input the radius.
3. Calculate Area = 3.142 × radius × radius.
4. Display the area.
5. Stop.

3. Algorithm to determine whether a student passed.

1. Start.
2. Input the student's score.
3. Check if the score is 50 or above.
4. If yes, display Pass; otherwise, display Fail.
5. Stop.

4. Algorithm to find the largest of three numbers.

1. Start.
2. Input three numbers A, B and C.
3. Compare the three numbers.
4. Identify the largest number.
5. Display the largest number.
6. Stop.

5. Algorithm to calculate the total cost of items purchased.

1. Start.
2. Input quantity.
3. Input unit price.
4. Calculate Total Cost = Quantity × Unit Price.
5. Display the total cost.
6. Stop.

Activity C — Pseudocode

1. Addition of two numbers

```
START
INPUT A
INPUT B
SUM = A + B
OUTPUT SUM
STOP
```

2. Determine whether a student passed or failed

```
START
INPUT score
IF score >= 50 THEN
    OUTPUT "PASS"
ELSE
    OUTPUT "FAIL"
END IF
STOP
```

Flowchart for addition: START → Input A and B → SUM = A + B → Display SUM → STOP.

Flowchart for pass/fail: START → Input score → Is score ≥ 50? → YES: Display PASS / NO: Display FAIL → STOP.

WEEK 2: DEVELOPMENT ENVIRONMENT AND FIRST JAVA PROGRAM

Practical 1 — First Java Program

Write a Java program that displays Hello World.

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("Welcome to COS 201.");
        System.out.println("I am learning Java programming.");
    }
}
```

Expected output

```
Hello, World!
Welcome to COS 201.
I am learning Java programming.
```

Practical 2 — Personalize Your Program

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("COURSE: COS 201");
        System.out.println("COURSE TITLE: Computer Programming I");
        System.out.println("PROGRAMME: Your Programme");
        System.out.println("WEEK: 2");
        System.out.println("TOPIC: Development Environment and Java");
        System.out.println("GOAL: To learn how to write Java programs.");
    }
}
```

Replace “Your Programme” with your actual programme.

Practical 3 — Predict the Output

```
public class PrintExample {
    public static void main(String[] args) {
        System.out.print("COS ");
        System.out.println("201");
        System.out.println("Java");
    }
}
```

Answer:

```
COS 201
Java
```

Practical 4 — Find and Correct the Error

The semicolon (;) is missing at the end of the print statement.

```
public class ErrorExample {
    public static void main(String[] args) {
        System.out.println("Learning Java");
    }
}
```

Practical 5 — README Documentation

```
# COS 201 – Computer Programming I

## Student Information
- Name: Your Full Name
- Matriculation Number: Your Matric Number
- Programme: Your Programme
- Course: COS 201
- Week: 2

## Practical Title
Development Environment and First Java Program
```

```
## Objective
To write, compile and run a simple Java program.

## Tools Used
- Java Development Kit (JDK)
- Visual Studio Code
- Git
- GitHub Desktop
- GitHub

## Files
- week02/HelloWorld.java

## How to Run
1. Open the terminal.
2. Compile using javac HelloWorld.java.
3. Run using java HelloWorld.

## Learning Outcome
I learned how to create, compile and run a Java program.
```

Practical 6 — Submission Message

Dear Lecturer,

I am submitting my Week 2 practical assignment for COS 201 – Computer Programming I.

Student Name: Your Full Name
Matriculation Number: Your Matric Number
Programme: Your Programme
Assignment: Development Environment and First Java Program
GitHub Repository: Paste your actual repository link.

Thank you.

WEEK 3: VARIABLES, DATA TYPES, EXPRESSIONS AND INPUT/OUTPUT

Practical 1 — Simple Calculator

Write a Java program that accepts two numbers and calculates their sum, difference, product and quotient.

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double firstNumber = input.nextDouble();

        System.out.print("Enter second number: ");
        double secondNumber = input.nextDouble();

        double sum = firstNumber + secondNumber;
        double difference = firstNumber - secondNumber;
        double product = firstNumber * secondNumber;

        System.out.println("Sum: " + sum);
        System.out.println("Difference: " + difference);
        System.out.println("Product: " + product);

        if (secondNumber != 0) {
            System.out.println("Quotient: " +
                (firstNumber / secondNumber));
        } else {
            System.out.println("Division is undefined.");
        }
    }
}
```

Sample output

```
Enter first number: 20
Enter second number: 4
Sum: 24.0
Difference: 16.0
Product: 80.0
Quotient: 5.0
```

Practical 2 — Student Average Calculator

Write a Java program that accepts a student's name and three examination scores, calculates the total and average, and displays the result.

```
import java.util.Scanner;

public class AverageCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter student name: ");
        String name = input.nextLine();

        System.out.print("Enter score for Course 1: ");
        double score1 = input.nextDouble();

        System.out.print("Enter score for Course 2: ");
        double score2 = input.nextDouble();

        System.out.print("Enter score for Course 3: ");
        double score3 = input.nextDouble();

        double total = score1 + score2 + score3;
        double average = total / 3.0;

        System.out.println("--- STUDENT RESULT ---");
        System.out.println("Student: " + name);
        System.out.println("Total: " + total);
```

```
        System.out.println("Average: " + average);
    }
}
```

Sample output

```
Enter student name: John
Enter score for Course 1: 70
Enter score for Course 2: 80
Enter score for Course 3: 90

--- STUDENT RESULT ---
Student: John
Total: 240.0
Average: 80.0
```

Practical 3 — Calculating the Total Cost of Items

Write a Java program that accepts an item name, quantity and unit price, then calculates the total cost.

```
import java.util.Scanner;

public class ItemCost {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter item name: ");
        String itemName = input.nextLine();

        System.out.print("Enter quantity: ");
        int quantity = input.nextInt();

        System.out.print("Enter unit price: ");
        double unitPrice = input.nextDouble();

        double totalCost = quantity * unitPrice;

        System.out.println("--- PURCHASE SUMMARY ---");
        System.out.println("Item: " + itemName);
        System.out.println("Quantity: " + quantity);
        System.out.println("Unit price: " + unitPrice);
        System.out.println("Total cost: " + totalCost);
    }
}
```

Sample output

```
Enter item name: Shoes
Enter quantity: 3
Enter unit price: 1500

--- PURCHASE SUMMARY ---
Item: Shoes
Quantity: 3
Unit price: 1500.0
Total cost: 4500.0
```

Practical 4 — README Documentation

```
## Week 3: Variables, Data Types,
Expressions and Input/Output

### Objectives
- Declare and initialize Java variables.
- Use appropriate data types.
- Perform arithmetic calculations.
- Read keyboard input using Scanner.
- Display formatted output.

### Programs
1. SimpleCalculator.java
   - Performs arithmetic calculations.
2. AverageCalculator.java
   - Calculates the total and average of three scores.
3. ItemCost.java
   - Calculates the total cost of purchased items.

### How to Run
```

```
javac SimpleCalculator.java
java SimpleCalculator

javac AverageCalculator.java
java AverageCalculator

javac ItemCost.java
java ItemCost
```

Practical 5 — Week 3 Submission Message

Dear Lecturer,

I am submitting my Week 3 practical assignment on Variables, Data Types, Expressions and Input/Output.

Student Name: Your Full Name
Matriculation Number: Your Matric Number
Programme: Your Programme
Course: COS 201 – Computer Programming I

Programs Submitted:
1. SimpleCalculator.java
2. AverageCalculator.java
3. ItemCost.java

GitHub Repository: Paste your actual repository link.

Thank you.

FINAL SUBMISSION CHECKLIST

Week	What to submit
Week 1	Activities A, B and C on paper.
Week 2	HelloWorld.java, README.md and GitHub repository.
Week 3	Three Java programs and updated README.md.

Important: Use one GitHub repository named COS201-Programming-I for all three weeks. Create separate folders named week01, week02 and week03. Replace sample student details with your actual information and test each Java program before submitting.

```
COS201-Programming-I/
|-- README.md
|-- week01/
|   `-- FirstProgram.java
|-- week02/
|   `-- HelloWorld.java
`-- week03/
    |-- SimpleCalculator.java
    |-- AverageCalculator.java
    `-- ItemCost.java
```

Note: The Week 1 activities and Week 2/3 practical answers are arranged from the available COS 201 lecture material. Confirm that these are the exact activities assigned by your lecturer before submission.